

UNIT 3

NETWORKING CONCEPTS – ADDRESSING AND PROTOCOLS



Learning Objectives

- Define the concept of network addressing.
- Differentiate between physical and logical addresses.
- Explain the structure and purpose of an IPv4 address.
- Understand the role of protocols in networking and describe key networking protocols.

Introduction to Networking Concepts

Networking concepts such as addressing and protocols are essential for ensuring smooth communication between devices. Network addressing assigns unique identifiers to devices, enabling them to locate and communicate with one another. Protocols, on the other hand, define the rules and procedures that govern data transmission across networks. These two concepts are the foundation of modern networking, enabling seamless interaction between millions of devices globally.

Network Addressing

Every device in a network requires a unique identifier to communicate effectively. This identifier, known as a network address, ensures that data reaches the correct destination. Network addressing can be categorized into two main types:

There are two types of addresses:

Physical Address (MAC Address)

- Given by the device manufacturer
- Found on the Network Interface Card (NIC)
- **Example:** 3C:52:82:4A:FE:2D

Logical Address (IP Address)

- Assigned by the network administrator or ISP
- Used to identify devices on different networks
- **Example:** 192.168.1.1

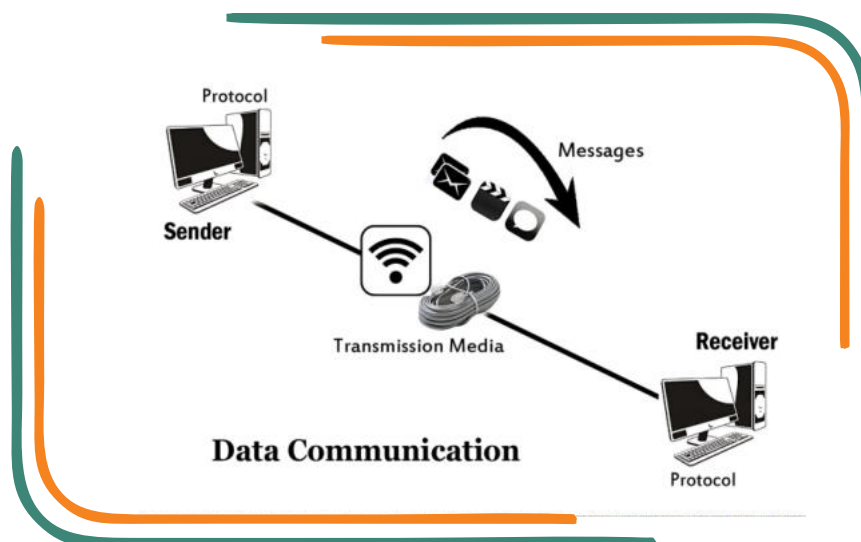
• Difference Between Physical and Logical Address

Physical Address	Logical Address
Attached to the NIC's ROM.	Assigned to a device by the network.
Refers to MAC (Media Access Control) Address.	Refers to IP (Internet Protocol) Address.
Permanent and globally unique.	Temporary and unique within the network.
48-bit address.	32-bit address in Ipv4.

How Do Computers Communicate in a Network?

When a computer sends a message:

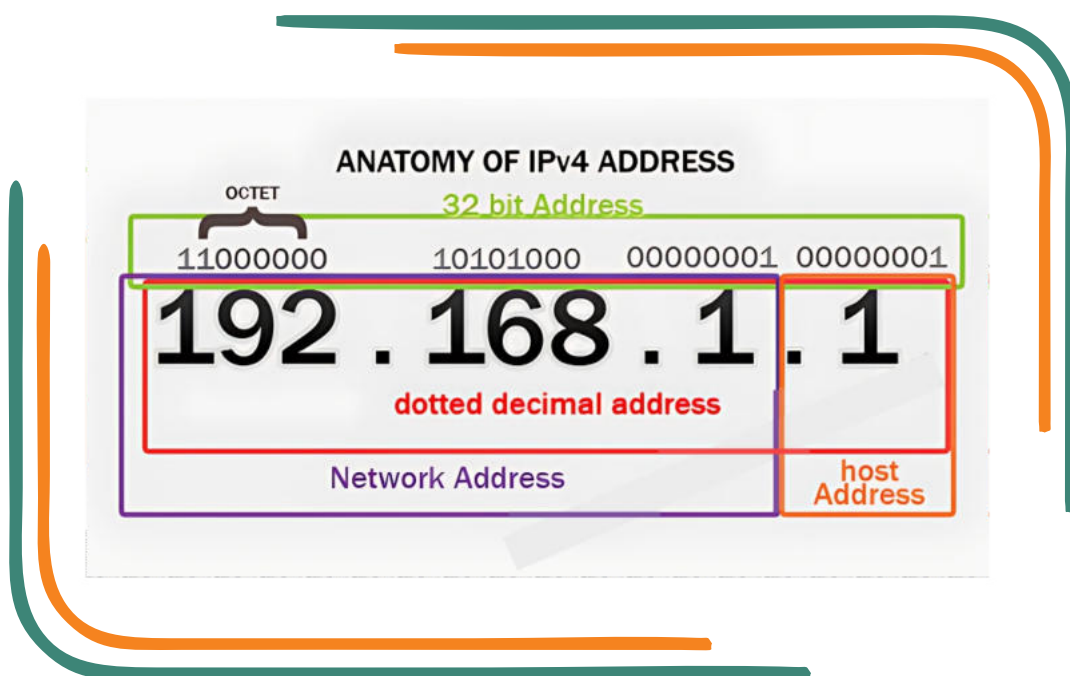
- It adds the IP address (logical address) of the receiver.
- The data is broken into packets.
- The data follows a protocol (e.g., TCP/IP).
- The receiver uses the IP and MAC addresses to accept the data.



IPv4 Address

An IPv4 address is a unique 32-bit number used to identify a device on a network. It is written in a dotted-decimal format, consisting of four segments separated by dots (e.g., 192.168.1.1). Each segment can range from 0 to 255. An IPv4 address is divided into two parts:

- **Network Portion:** Identifies the network to which the device belongs.
- **Host Portion:** Identifies the specific device within the network.



Networking Protocols

Protocols are standardized rules and conventions that define how data is transmitted, received, and interpreted over a network. They ensure reliable communication and interoperability between different devices and systems.

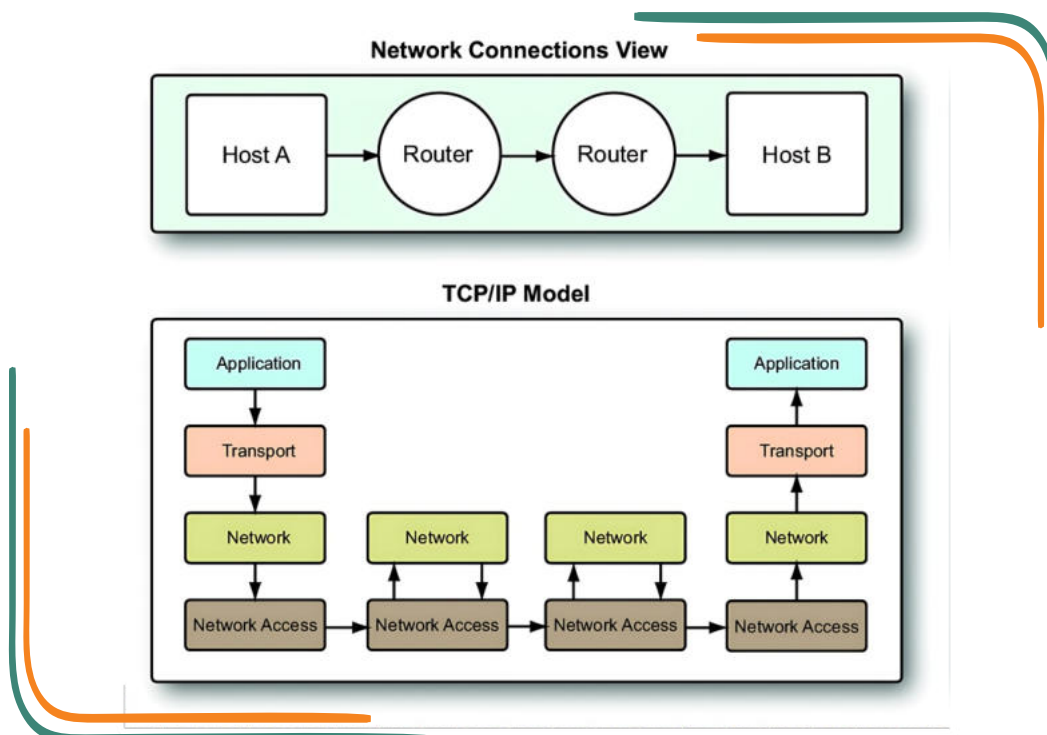
Key Networking Protocols

• TCP/IP (Transmission Control Protocol/Internet Protocol)

TCP/IP is the foundational protocol suite for the internet.

It consists of two main components:

- TCP: Ensures reliable delivery of data by breaking it into packets and reassembling them at the destination.
- IP: Handles the addressing and routing of packets to ensure they reach the correct destination.



• HTTP/HTTPS (Hypertext Transfer Protocol/Secure)

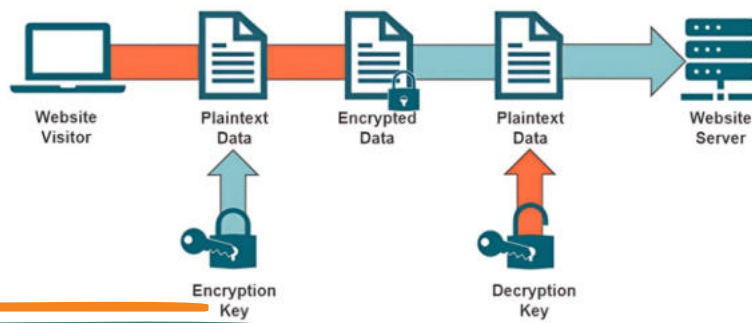
HTTP is the protocol used for transferring web pages over the internet. HTTPS adds an extra layer of security by encrypting the data being transmitted.

- **HTTP:** Transmits data in plaintext.
- **HTTPS:** Uses SSL/TLS encryption to secure data.

How Insecure Website Communications Work (HTTP)



How Secure Website Communications Work (HTTPS)

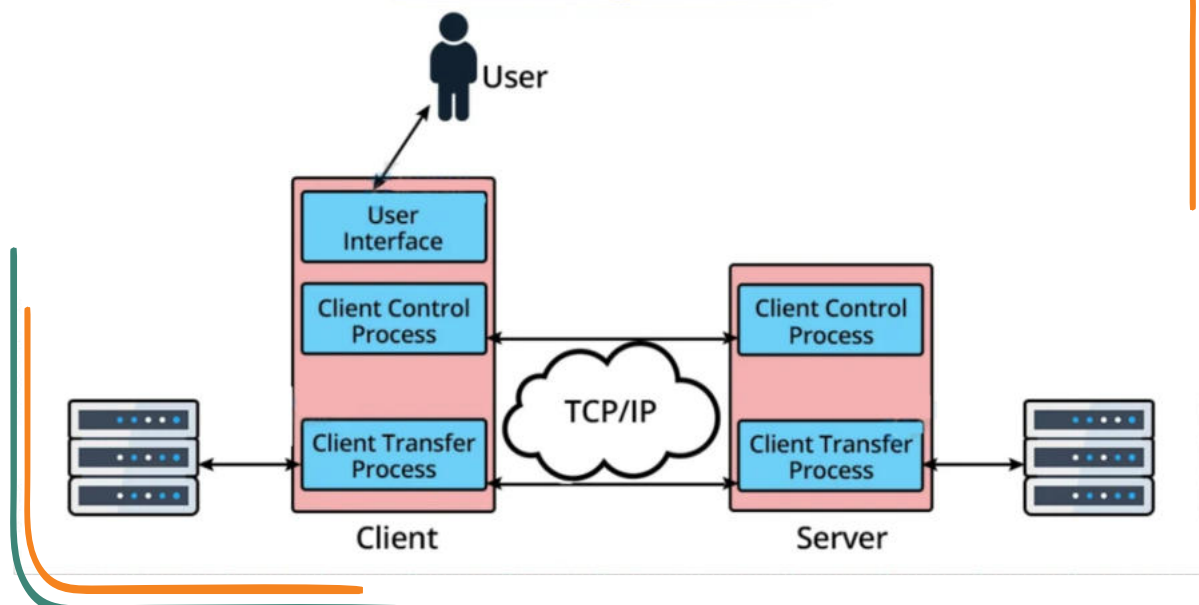


• FTP (File Transfer Protocol)

FTP is used for transferring files between computers over a network. It supports both uploading and downloading files.

- **Advantages:** Simple and efficient file transfers.
- **Disadvantages:** Data is not encrypted, making it less secure.

Working of FTP

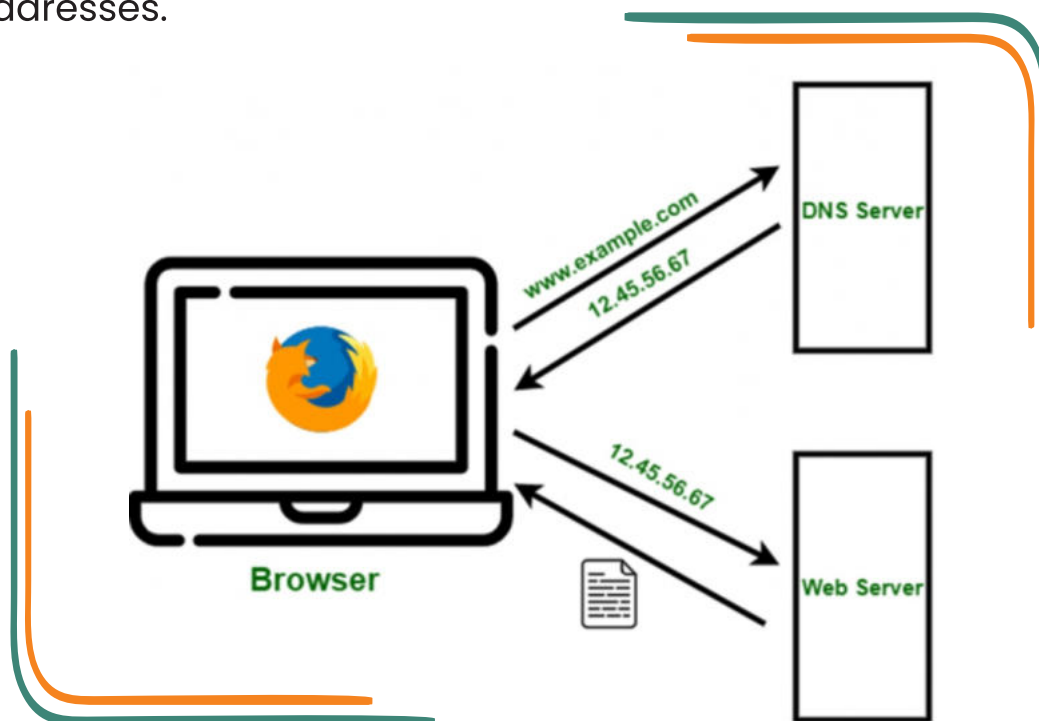


• DNS (Domain Name System)

DNS translates human-readable domain names (e.g., `www.example.com`) into IP addresses that computers use to identify each other on a network.

• **Function:** Simplifies internet navigation for users.

• **Importance:** Eliminates the need to memorize complex IP addresses.



Understanding network addressing and protocols is essential for anyone studying computer networking. Addressing ensures that devices can identify and communicate with each other, while protocols define the rules that govern these interactions. Together, they form the backbone of modern networking, enabling the seamless exchange of data across the globe.

These foundational concepts pave the way for more advanced topics in networking and internet technologies.

EXERCISES

01: Choose the correct answer.

1.What is the main purpose of network addressing?

- a) To secure data transmission
- b) To identify devices in a network
- c) To improve network speed
- d) To store network data

2.What type of address is associated with the NIC's ROM?

- a) IPv4 address
- b) Logical address
- c) Physical address
- d) Domain name

3.Which of the following is a characteristic of a logical address?

- a) Permanent and globally unique
- b) Refers to the MAC address
- c) Assigned to a device by the network
- d) 48-bit address

4.How many bits are used in an IPv4 address?

- a) 16 bits
 - b) 32 bits
 - c) 48 bits
 - d) 64 bits
-

5. What does the TCP/IP protocol suite handle?

- a) Data encryption
 - b) Data addressing and routing
 - c) Data compression
 - d) Data analysis
-

6. What is the function of DNS?

- a) Transfers files between computers
 - b) Converts domain names to IP addresses
 - c) Encrypts data for secure transmission
 - d) Splits data into packets
-

7. Which protocol uses encryption to secure data during transfer?

- a) FTP
 - b) HTTP
 - c) DNS
 - d) HTTPS
-

8. What is a disadvantage of FTP?

- a) It is slow
 - b) Data is not encrypted
 - c) It is difficult to use
 - d) It requires a secure connection
-

9. In IPv4, which part identifies the specific device within the network?

- a) Network portion
- b) Host portion
- c) MAC address
- d) DNS

10. Which of the following is the primary use of HTTP?

- a) Transmitting web pages
 - b) Transferring files
 - c) Resolving domain names
 - d) Routing data packets
-

02: Fill in the blanks.

1. A _____ address is a permanent, globally unique identifier attached to the NIC's ROM.
2. The two main parts of an IPv4 address are the _____ portion and the _____ portion.
3. _____ is a protocol used for transferring web pages over the internet.
4. The _____ protocol ensures secure transmission by encrypting data.
5. _____ is the protocol that translates human-readable domain names into IP addresses.
6. An IPv4 address is written in a _____-decimal format.
7. _____ ensures reliable delivery of data by breaking it into packets.
8. A _____ address is assigned to a device by the network and is temporary.
9. The main disadvantage of _____ is that it does not encrypt data.
10. The TCP/IP protocol suite consists of two main components: _____ and _____.

03: Write True or False for each statement.

	TRUE	FALSE
1. An IPv4 address is a 64-bit address.	☐	☐
2. HTTP and HTTPS are two types of networking protocols used for web page transmission.	☐	☐
3. The MAC address is a logical address.	☐	☐
4. DNS helps users navigate the internet by converting domain names to IP addresses.	☐	☐
5. The host portion of an IPv4 address identifies the network to which the device belongs.	☐	☐
6. TCP is responsible for ensuring reliable delivery of data.	☐	☐
7. FTP is a protocol used for file transfer and encrypts the transferred data.	☐	☐
8. A physical address is temporary and unique only within a network.	☐	☐
9. HTTPS adds an extra layer of security by using SSL/TLS encryption.	☐	☐
10. DNS is not required for navigating the internet.	☐	☐

04: Answer the following questions briefly:

1. What is the difference between physical and logical address?
2. Name any three networking protocols and their uses.
3. Explain how IPv4 addressing works.
4. Why are protocols important in networking?
5. What is the difference between HTTP and HTTPS?
6. What are the advantages and disadvantages of using FTP for file transfers?

05: Provide detailed answers of the following questions:

1. How does a computer identify another computer in a network?
2. How many numbers are there in an IPv4 address, and what is the range of each number?
3. What is a protocol in networking?
4. Why are protocols necessary in a computer network?

Lab Activity : FTP File Transfer



- Set up an FTP server and client on two different computers.
- Transfer a file from one computer to another using the FTP protocol.
- Observe and report on the file transfer process and the security limitations of FTP.